

IIIT Hyderabad Invites Applications for Ph.D. Opportunities in the Software Engineering Research Centre (SERC)

Application End Date: 15-09-2026

The Software Engineering Research Centre ([SERC](#)) at IIIT Hyderabad is a premier research centre dedicated to advancing the field of software engineering through innovative research, education, and industry collaboration. SERC conducts cutting-edge research in areas including software architecture, software testing, AI-driven software engineering, cybersecurity, and developer productivity. The centre actively collaborates with academia, industry, and government organizations to address real-world software challenges. Through its Ph.D. programs, research initiatives, and technology transfer activities, SERC contributes significantly to the advancement of the software engineering ecosystem in India and beyond.

Ph.D. Opportunities in the Following Research Areas:

◆ Human-Centred Immersive Healthcare Experiences

Research at the intersection of Human-Computer Interaction (HCI) and Human-Centred Design (HCD), leveraging AI, sensors, wearables, AR/VR/MR/XR, and health data to develop personalized and immersive digital healthcare systems.

◆ Trustworthy and Accessible Digital Systems for Digital Living & Wellbeing

Research in Human-Centred AI, HCI, and Design Thinking to create inclusive, ethical, privacy-aware, and trustworthy digital systems that promote digital well-being and socially impactful innovation.

◆ AI and Educational Technology

Application of AI and Large Language Models (LLMs) to develop Virtual Labs and supporting cloud infrastructure for scalable and interactive learning environments.

◆ Computer Science Education

Research on methodologies, tools, and technologies for improving Computer Science education and learning experiences.

◆ Trustworthy Parallel Computing

Semantics and verification of concurrent programs under weak memory models, including model checking, controlled testing, and fuzzing techniques for reliable parallel systems.

◆ AI for Software Engineering and Theorem Proving

LLM-based translation of natural language to formal specifications, property-based testing, and ML-driven automation of proof assistants through tactic generation, proof search, and lemma synthesis.

◆ Software Testing for Modern Systems:

Conformance-based interaction testing, automated test-case generation, and validation pipelines for AI-enabled applications

◆ Code Smell Detection and Quality:

Improving correctness of semantic smell-detection tools and identifying emerging code smells in AI-generated and AI-based systems

◆ Engineering Agentic AI Systems

Architectural approaches for Agentic AI systems, Sustainable Agentic AI, AgentOps, evaluation frameworks, and Agentic Memory.

◆ Agentic/Autonomous Software Engineering

Architecture Knowledge Management, Run-time Adaptation, Sustainable Coding Agents, and Sovereign Coding Agents.

We are seeking highly motivated students with a strong mathematical background. In particular, a solid understanding of probability theory is essential. Prior exposure to statistics, optimization, machine learning, formal methods, or software engineering research will be an added advantage.

Eligibility: B.E/B.Tech/M.E/M.Tech from CSE/ECE

Selection Process:

Candidates shortlisted after initial screening shall be asked to spend 3–4 months working on a research problem or internship-style project before the formal admission process is initiated. This process helps both the student and the advisor evaluate research fitness and readiness for Ph.D. work. Formal admission will be through the Standing Committee route at IIIT Hyderabad.

Interested candidates should submit a Cover Letter, Curriculum Vitae (CV), and Statement of Purpose (SOP) to serc.admin@iiit.ac.in, clearly mentioning the specific research area(s) they are interested in.

Applications will remain open until September 15, 2026. Applications will be reviewed every two weeks until the closing date; therefore, interested candidates are encouraged to apply early.